Extracting the second-order residual field at the particle

A toy-model test of Taylor continuation from the contour

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Abstract

The contour method gives the second-order field on a complex contour inside the world-tube and on the real line outside it. That is enough for fluxes, but a local self-force also needs the second-order residual field, and its gradient, at the particle. We test whether this can be obtained by subtracting a continued second-order puncture from the contour solution, where both are of order one, and continuing the difference analytically from the contour to r_0 by a Taylor series generated from the radial ODE. In the flat-space toy model with $Q = \frac{1}{2}\partial_Z^2(AB)$ and the exact second-order singular field, the continued residual is analytic at r_0. The Taylor series converges geometrically, with the ratio set by the distance to the next singularity at $z = 0$, and recovers every residual mode and its radial derivative at r_0 to 10^{-32}. The mode sums for $\Phi_2^{\rm R}(\mathbf{x}_0)$ and $\partial_r\Phi_2^{\rm R}(\mathbf{x}_0)$ are then limited only by ℓ-truncation. With deliberately truncated punctures the continued residual has a branch point at r_0, and convergence becomes algebraic: $N^{-(k+2)}$ for a puncture error $\propto \rho^k$, two orders better than the 3D smoothness suggests. A simpler baseline, reading the residual at $r_0 + i\eta$ on lower contours and extrapolating $\eta \to 0$, also works, without the cancellation we had expected. The toy is static, so it tests the extraction of the local residual but not the split into dissipative and conservative parts.

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1 Question and summary

The contour method solves for the full second-order field. Inside the worldtube the modes are known on the contour $z = y(a)$; outside, on the real line. The fluxes at the boundaries follow directly. A local self-force needs more: the regular (residual) part of the field and its gradient at the particle, $\Phi_2^{\rm R}(\mathbf{x}_0)$ and $\nabla\Phi_2^{\rm R}(\mathbf{x}_0)$. In mode language, these are the residual modes $\phi_\ell^{\rm R}(r_0)$ and $\partial_r\phi_\ell^{\rm R}(r_0)$, summed over ℓ.

The proposal tested here has two steps.

1. On the contour, subtract the continued second-order puncture from the contour solution, $\tilde\psi_\ell^{\rm R}(z) = \tilde\psi_\ell(z) - \tilde\phi_\ell^{\rm P}(z)$. On the contour $|z - r_0| \geq 0.298$, so both terms are of order one and no singular quantity is subtracted from another.

2. Continue $\tilde\psi_\ell^{\rm R}$ from the contour to r_0 with a Taylor series, and take its real part.

Summary of results.

- *Exact puncture:* the continued residual is analytic at r_0 (its monodromy around r_0 is below 4×10^{-20}). The Taylor series about the apex $r_0 + i$ converges geometrically with ratio 0.199, against the predicted $|r_0 - z_e|/|z_e| = 0.196$. It recovers $\phi_\ell^{\rm R}(r_0)$ and $\partial_r\phi_\ell^{\rm R}(r_0)$ for all $\ell \leq 20$ to better than 10^{-34} and 10^{-32}, reaching 10^{-20} after 29–50 terms.

- *Mode sums:* $\Phi_2^{\rm R}(\mathbf{x}_0)$ and $\partial_r\Phi_2^{\rm R}(\mathbf{x}_0)$ have errors 9.6×10^{-11} and 4.1×10^{-10} at $\ell_{\rm max} = 20$, entirely from ℓ-truncation.

- *Truncated punctures* $\Phi_2^{\rm P} = \Phi_2^{\rm S} + \epsilon q^2\rho^k$: the convergence is algebraic, with value errors $\propto N^{-(k+2)}$ (fitted slopes -3.0, -5.1, -7.3 for $k = 1, 3, 5$).

- *Baseline:* reading the residual at $r_0 + i\eta$ has a bias $-\eta\,\mathrm{Im}\,\tilde\psi_\ell^{\rm R\prime}(r_0)$; Richardson extrapolation over $\eta = 1, \ldots, 1/128$ recovers the exact case to 3×10^{-15}–5×10^{-11}.

- *Controls:* continuing the puncture one-sidedly (dropping its kink at r_0) fails, as it should. Blending the finished, kinked puncture modes with H works, contrary to our prediction (Sec. 4.6).

2 Set-up

2.1 Toy model and contour

The first-order field satisfies $\nabla^2\Phi_1 = -4\pi q\,\delta^3(\mathbf{x} - \mathbf{x}_0)$, with

$$\Phi_1 = \frac{q}{\rho} + \Phi_1^{\rm R}, \qquad \Phi_1^{\rm R} = c_0 + c_1 Z = A + c_1 d, \qquad A = c_0 + c_1 r_0, \tag{1}$$

where $Z = r\mu$ is the Cartesian coordinate along the axis, $\mathbf{x}_0 = r_0\hat{Z}$, and $d = Z - r_0$. The second-order field satisfies $\nabla^2\Phi_2 = \frac{1}{2}\partial_Z^2(\Phi_1^2)$. Its exact solution is $\Phi_2 = \frac{1}{2}\partial_Z^2\chi$, with $\nabla^2\chi = \Phi_1^2$. The parameters are $q = 1$, $r_0 = 5$, $c_0 = 1$, $c_1 = 1/10$, on the domain $r \in [1, 10]$. The contour is $y(a) = a + i\big(1 - 4(a - r_0)^2\big)^6$ on $[a_L, a_R] = [4.5, 5.5]$, with apex $r_0 + i$. The radial operator is $\mathcal{L}_\ell\psi = \psi'' + 2\psi'/r - \ell(\ell+1)\psi/r^2$.

2.2 The exact second-order singular field

Write $\Phi_1^2 = q^2/\rho^2 + 2qA/\rho + 2qc_1 d/\rho + (\Phi_1^{\rm R})^2$. Using $\nabla^2\ln\rho = 1/\rho^2$, $\nabla^2\rho = 2/\rho$ and $\nabla^2(d\rho) = 4d/\rho$, the singular part of χ is

$$\chi^{\rm S} = q^2\ln\rho + qA\rho + \tfrac{1}{2}qc_1\,d\rho, \qquad \Phi_2^{\rm S} = \tfrac{1}{2}\partial_Z^2\chi^{\rm S}, \tag{2}$$

with the derivative taken in the distributional sense. Then $\Phi_2^{\mathrm{R}} = \Phi_2 - \Phi_2^{\mathrm{S}} = \frac{1}{2}\partial_Z^2(\chi - \chi^{\mathrm{S}})$ is smooth at $\mathbf{x}_0$. Because Φ_1^{R} is linear, the singular field is a finite closed form. In GR, and in any toy with a non-polynomial regular field, it is an infinite local series that must be truncated.

2.3 An injected regular field

With the boundary data used in earlier stages, the SS and SR solutions equal $\frac{1}{2}\partial_Z^2$ of the singular parts of χ exactly. The residual from SS and SR would then vanish identically, and the test would be trivial. To give every ℓ a nonzero residual, the agent added a harmonic field through the Cauchy data at a_L:

$$h = \frac{B_o}{|\mathbf{x} - r_o\hat{Z}|} + \frac{B_i}{|\mathbf{x} - r_i\hat{Z}|}, \qquad B_o = 10,\ r_o = 15,\ B_i = 1,\ r_i = \tfrac{1}{2}, \tag{3}$$

which is harmonic on the domain. Its modes are $h_\ell(r) = \sqrt{4\pi/(2\ell+1)}\,[B_o r^\ell/r_o^{\ell+1} + B_i r_i^\ell/r^{\ell+1}]$. The RR piece is solved on the real line, as before, and is regular at r_0. The exact residual values at the particle are

$$\Phi_2^{\mathrm{R}}(\mathbf{x}_0) = \tfrac{1}{6}c_0^2 + \tfrac{3}{5}c_0c_1r_0 + \tfrac{1}{2}c_1^2r_0^2 + h(\mathbf{x}_0) = 1.81388888888888888889, \tag{4}$$

$$\partial_r\Phi_2^{\mathrm{R}}(\mathbf{x}_0) = 0.16061728395061728395. \tag{5}$$

On the axis $\partial_r = \partial_Z$, and by axisymmetry this is the whole gradient.

3 Method

3.1 The contour solution and the continued puncture

SS and SR are solved together on the contour by variation of parameters, with 60 Gauss panels of 20 nodes in a and closed-form sources evaluated at 50 digits. This agrees with the NDSolve contour solve of the earlier stages to 10^{-15}. The continued sources are those of Stage 9: $\tilde{s}_{\mathrm{SS}}$ with $2\log(z + r_0) - 2\log(z - r_0)$, and $\tilde{s}_{\mathrm{SR}}$ through the $\Theta \to H$ continuation of the coupling kernels.

The continued puncture $\tilde{\phi}_\ell^{\mathrm{S}}$ is built in the same way as the source:

- *SS part:* the modes of $\frac{1}{2}\partial_Z^2(q^2 \ln\rho)$ are the same real function on both sides of r_0, with $\ln|r - r_0|$. They are continued by $\ln|r - r_0| \to \log(z - r_0)$.
- *SR part:* the undifferentiated modes $F_j = [\chi_{\mathrm{SR}}]_j$ are C^2 at r_0. They are continued by the blend $F_{\mathrm{in}} + (F_{\mathrm{out}} - F_{\mathrm{in}})H$, with $H = 1 + \frac{i}{\pi}\log(z - r_0)$, and then the $\frac{1}{2}D_z^2$ ladder is applied.

Three identities were checked:

- $\tilde{s}_{\mathrm{SS}} = \mathcal{L}_\ell\tilde{\phi}_{\mathrm{SS}}^{\mathrm{S}}$ exactly (10^{-40});
- $\mathrm{Re}\,\tilde{\phi}_\ell^{\mathrm{S}}(r + i0)$ equals the modes of Φ_2^{S} on both sides of r_0;
- the residual source $\tilde{s}^{\mathrm{R}} = \tilde{s}_{\mathrm{SS}} + \tilde{s}_{\mathrm{SR}} - \mathcal{L}_\ell\tilde{\phi}_\ell^{\mathrm{S}}$ is imaginary on the axis and bounded at r_0.

The last of these is expected: the physical residual source of SS and SR is zero, so only a shadow is left.

3.2 Why the residual should be analytic at r_0

$\tilde{\psi}^{\mathrm{R}}_\ell$ is analytic between the contour and the real axis. Its real part on the axis is the physical residual mode, which for the exact puncture is real-analytic at r_0 and contains no δ terms. Subtracting the analytic continuation of that real part leaves a function whose real part vanishes on the axis. By Schwarz reflection, such a function extends analytically across the axis, including through r_0. So $\tilde{\psi}^{\mathrm{R}}_\ell$ should be analytic in a disc around r_0, and a Taylor series about a contour point z_e should converge at r_0 with ratio $|r_0 - z_e|/|z_e|$, set by the regular singular point of $\mathcal{L}_\ell$ at $z = 0$.

With a truncated puncture, the physical residual is only finitely differentiable at r_0. Its continuation then contains a term like $(z - r_0)^m \log(z - r_0)$, so r_0 is a branch point on the edge of every Taylor disc and convergence can only be algebraic.

3.3 Taylor coefficients from the ODE

Fitting Taylor coefficients to values on the contour would be numerical continuation off a curve, which is exponentially ill-conditioned. Instead the coefficients are generated by the ODE itself. With $z = z_e + w$ and $\tilde{\psi}^{\mathrm{R}}_\ell = \sum_n p_n w^n$, the equation $z^2\psi'' + 2z\psi' - \ell(\ell+1)\psi = z^2\tilde{s}^{\mathrm{R}}$ gives

$$p_{n+2} = \frac{\sigma_n - 2z_e(n+1)^2 p_{n+1} - \big[n(n+1) - \ell(\ell+1)\big]p_n}{z_e^2(n+2)(n+1)}, \tag{6}$$

where σ_n are the Taylor coefficients of $z^2\tilde{s}^{\mathrm{R}}$ about z_e. The starting values $p_0 = \tilde{\psi}^{\mathrm{R}}_\ell(z_e)$ and $p_1 = \tilde{\psi}^{\mathrm{R}\prime}_\ell(z_e)$ come from the contour solve. The σ_n come from a Cauchy integral of the source on the circle $|z - z_e| = 0.7|z_e - r_0|$, using the trapezoidal rule with 256 points. Every quantity involved is of order one. The partial sums at r_0 are

$$S_N = \sum_{n\le N} p_n (r_0 - z_e)^n, \qquad S'_N = \sum_{n\le N} n\, p_n (r_0 - z_e)^{n-1}. \tag{7}$$

Two expansion points were used: the apex $a_e = 5$ ($|z_e - r_0| = 1$) and $a_e = 5.2$ ($|z_e - r_0| = 0.404$). As a cross-check, the residual ODE was also solved around a circle $|z - z_e| = r_c$ and the coefficients taken by Cauchy integrals. These agree with the recursion to 2×10^{-19} in the coefficients and 6×10^{-16} in the partial sums at $N = 40$.

4 Results

4.1 The residual on the contour

Figure 1 shows $\tilde{\psi}^{\mathrm{R}}_\ell$ along the contour for several ℓ. Its real part at a_L and a_R equals the physical residual there, as required. Its imaginary part, the shadow, is of order one and grows with ℓ: $\mathrm{Im}\,\tilde{\psi}^{\mathrm{R}}_\ell(a_R)$ reaches -11 at $\ell = 20$. Since $\phi^{\mathrm{R}}_\ell(r_0) = 1.1 \times 10^{-10}$ at $\ell = 20$, the extraction recovers a small real part beside a large imaginary one. At the 50 digits used here this costs nothing, but in double precision the absolute accuracy of a high-ℓ mode would be set by the size of the shadow, not by the size of the residual.

4.2 Exact puncture: geometric convergence

Table 1 lists the extracted residual modes. For every $\ell \le 20$ the real part of the Taylor sum at r_0 equals the exact $h_\ell(r_0)$ to better than 10^{-34}, and the radial derivative to better than 10^{-32}, which is the floor of the 50-digit arithmetic. A fit $|p_n(r_0 - z_e)^n| \sim n^c \varrho^n$ to the coefficients gives $\varrho = 0.199$ at the apex and 0.080 at $a_e = 5.2$, against the predicted 0.196 and 0.078. The monodromy of $\tilde{\psi}^{\mathrm{R}}_\ell$ around r_0 is below 4×10^{-20}, so r_0 is not a singular point. Every contour point works, and points closer to r_0 converge faster.

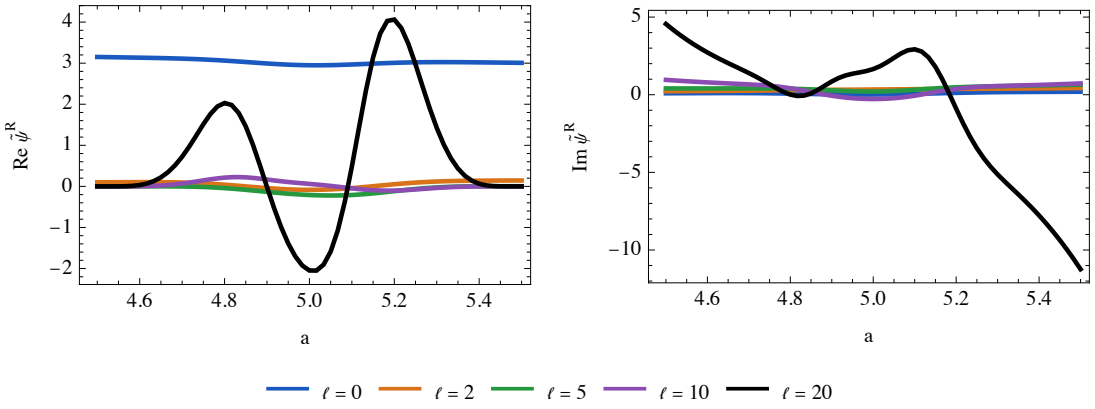


Figure 1: The residual $\tilde{\psi}^{\mathrm{R}}_\ell = \tilde{\psi}_\ell - \tilde{\phi}^{\mathrm{S}}_\ell$ along the contour, exact puncture. Left: real part; right: imaginary part (the shadow).

Table 1: Exact puncture: residual modes at r_0 from the Taylor series about the apex ($N = 120$). N_{20} is the number of terms needed for 10^{-20}.

ℓ	$\mathrm{Re}\, S_N = h_\ell(r_0)$	value error	derivative error	$\mathrm{Im}\, \tilde{\psi}^{\mathrm{R}}_\ell(r_0)$	N_{20}
0	3.0722533415696	$< 10^{-36}$	$< 10^{-34}$	0.144	29
2	0.12060258176715	$< 10^{-36}$	$< 10^{-34}$	0.320	32
5	$2.9344555955148 \times 10^{-3}$	$< 10^{-35}$	$< 10^{-33}$	0.455	34
10	$8.7335788933812 \times 10^{-6}$	$< 10^{-35}$	$< 10^{-32}$	0.505	39
15	$2.9581078715916 \times 10^{-8}$	$< 10^{-34}$	$< 10^{-32}$	0.184	46
20	$1.0585141067450 \times 10^{-10}$	$< 10^{-34}$	$< 10^{-32}$	-1.171	50

The value $\mathrm{Im}\, \tilde{\psi}^{\mathrm{R}}_\ell(r_0)$ is of order one: the shadow is carried to r_0, but only the real part is physical.

4.3 Truncated punctures: algebraic convergence

To mimic GR, where punctures are always truncated, the puncture was replaced by $\Phi_2^{\mathrm{P}} = \Phi_2^{\mathrm{S}} + \epsilon q^2 \rho^k$ with odd $k = 1, 3, 5$ and $\epsilon = r_0^{-(k+2)}$. The 3D residual is then C^{k-1}, and its exact value at $\mathbf{x}_0$ is unchanged because ρ^k vanishes there. This is the second-order analogue of the $cq\rho^n$ trick used for first-order punctures in Stage 6.

Figure 2 and Table 2 show the result. Convergence is algebraic, as predicted, but with the exponent $k + 2$, not k:

- The mode functions $[\rho^k]_\ell$ are C^{k+1} at r_0, two orders smoother than the 3D field. The continued residual therefore has a $(z - r_0)^{k+2} \log(z - r_0)$ branch point, and the value at r_0 converges as $N^{-(k+2)}$.
- Off the vertical ($a_e = 5.2$) the derivative converges one order more slowly, as $N^{-(k+1)}$.
- At the apex, which lies directly above r_0, the derivative converges as fast as the value. There the leading term $(i/\pi)\, j\, (z - r_0)^{k+2} \log(z - r_0)$ has Taylor terms of constant phase, which feed only the real part of the value and the imaginary part of the derivative.

Moving the expansion point closer to r_0 reduces the prefactor, by $|z_e - r_0|^{k+2}$, but not the exponent.

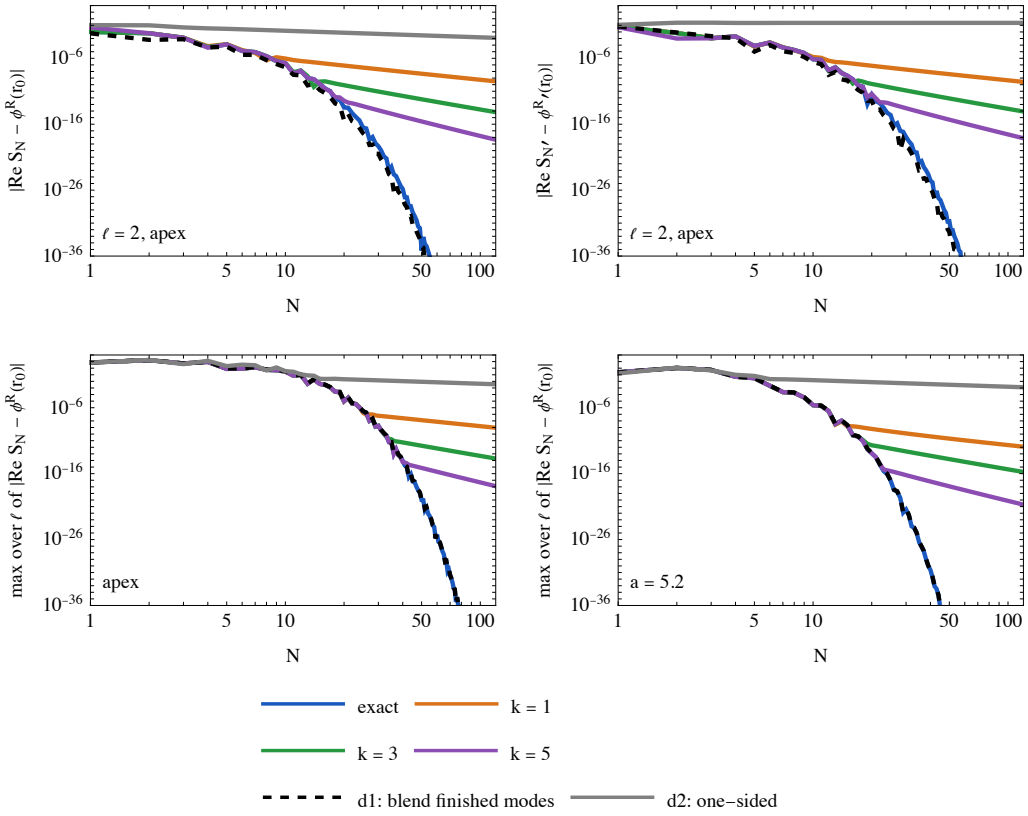


Figure 2: Error of the Taylor partial sums at r_0 against the number of terms N. Top: $\ell = 2$ about the apex, value (left) and radial derivative (right). Bottom: maximum over $\ell \leq 20$, about the apex (left) and $a_e = 5.2$ (right). The controls d1 and d2 are described in Sec. 4.6.

4.4 Mode sums at the particle

Summing the extracted modes, with the RR contribution taken from the real line, gives the residual at the particle (Table 3, Fig. 3). In every case the error equals the error of the same sum built from the exact modes, so the Taylor step contributes nothing visible.

- For the exact puncture, the ℓ-truncation error comes from h and falls as $(r_0/r_o)^\ell = 3^{-\ell}$.
- For the truncated punctures, it comes from the slowly convergent mode sum of ρ^k at $\mathbf{x}_0$, as in any puncture scheme of that order.

4.5 Baseline: lower contours and $\eta \to 0$

A simpler route is to use contours of decreasing apex height η, $y_\eta(a) = a + i\eta(1-4(a-r_0)^2)^6$, and read $\tilde{\psi}_\ell - \tilde{\phi}^{\rm P}_\ell$ at $r_0 + i\eta$. Figure 4 shows that the real part has a bias linear in η, with coefficient $-\operatorname{Im}\tilde{\psi}^{\rm R\prime}_\ell(r_0)$. This is the shadow's derivative: $\operatorname{Re}\tilde{\psi}^{\rm R}_\ell(r_0+i\eta) = \operatorname{Re}\tilde{\psi}^{\rm R}_\ell(r_0) - \eta \operatorname{Im}\tilde{\psi}^{\rm R\prime}_\ell(r_0) + O(\eta^2)$. At $\eta = 1/128$ the bias is still 1.6×10^{-3} for $\ell = 2$.

Richardson extrapolation over the eight values $\eta = 1, \ldots, 1/128$ recovers the exact case to 3×10^{-15}–5×10^{-11} for $\ell \leq 10$. For truncated punctures it saturates at the $\eta^{k+2}\log\eta$ level (10^{-10} for $k = 1$).

We had expected this route to lose digits to cancellation between two singular quantities. It does not: the field modes are only logarithmically singular at r_0, so $|\tilde{\psi}_\ell|$ and $|\tilde{\phi}^{\rm S}_\ell|$ at the apex stay of order one. The cost is elsewhere: the source grows like η^{-2} (to 1.3×10^3 at $\eta = 1/128$), and resolving it needs more quadrature nodes (840 at $\eta = 1$, 1120 at $\eta = 1/128$).

Table 2: Convergence of the Taylor sums at r_0: median fitted slopes of the real-part errors against N, and the maximum error over $\ell \le 20$ at $N = 120$ (apex).

		slope against N		max error, $N = 120$	
case	value, apex	deriv., apex	deriv., $a_e = 5.2$	value	derivative
exact	0.199^N	geometric	0.080^N	$< 10^{-34}$	$< 10^{-32}$
$k = 1$	-3.04	-3.08	-2.04	8.7×10^{-10}	7.1×10^{-10}
$k = 3$	-5.14	-5.21	-4.16	1.8×10^{-14}	2.2×10^{-14}
$k = 5$	-7.31	-7.42	-6.55	1.1×10^{-18}	1.9×10^{-18}

Table 3: Mode sums at $\mathbf{x}_0$ with $\ell_{\max} = 20$. The errors equal those of the same sums built from exact modes.

case	$\Phi_2^{\mathrm{R}}(\mathbf{x}_0)$	error	$\partial_r \Phi_2^{\mathrm{R}}(\mathbf{x}_0)$	error
exact	1.81388888879329	9.6×10^{-11}	0.16061728353954	4.1×10^{-10}
$k = 1$	1.81198302958710	1.9×10^{-3}	0.16042671314077	1.9×10^{-4}
$k = 3$	1.81390192019408	1.3×10^{-5}	0.16062119295924	3.9×10^{-6}
$k = 5$	1.81388843917625	4.5×10^{-7}	0.16061705873102	2.3×10^{-7}
exact value	1.81388888888889		0.16061728395062	

4.6 Controls: how the puncture is continued

Two alternative continuations of the puncture were tried.

- *d1: H-blend of the finished modes.* The finished, differentiated SR puncture modes P are kinked (C^0) at r_0. Blending them directly, $P_{\mathrm{in}} + (P_{\mathrm{out}} - P_{\mathrm{in}})H$, gives the same real values as blending first and differentiating afterwards: the error is below 10^{-34}. We had predicted that this would fail, by analogy with the source (PRL Appendix C, Step 6), where blending the finished source loses its δ terms. The difference is that the puncture is a field. For a kinked function, $\mathcal{L}_\ell\big[P_{\mathrm{in}} + (P_{\mathrm{out}} - P_{\mathrm{in}})H\big]$ contains $[P']\,\delta_{\mathrm{ext}}$ automatically, so the kink reproduces the δ in $\mathcal{L}_\ell P$. The two continuations differ by a function that is imaginary on the axis and vanishes at r_0, a harmless shadow. The finished SR source, by contrast, contains δ and δ' terms that no blend of functions can represent.

- *d2: one-sided continuation.* Continuing P_{out} alone drops the kink, and hence the δ in $\mathcal{L}_\ell P$. This fails, as it should. Its real part is already wrong at a_L, by the jump times $\Theta(r_0 - r)$. The residual acquires a term $(i/\pi)c_\ell(z - r_0)\log(z - r_0)$, where c_ℓ is the dropped δ coefficient. The value then converges only as $1/N$, and the derivative tends to the mean of the two one-sided derivatives, with error $|c_\ell|/2$ (0.11 for $\ell = 0$, 0.24 for $\ell = 2$). The mode sum for $\partial_r \Phi_2^{\mathrm{R}}(\mathbf{x}_0)$ is wrong by 13.

The requirement on the puncture is therefore the first consistency condition of the earlier note: the real part of the continued puncture on the axis must equal the modes of the 3D puncture on both sides of r_0. Keeping the kink is enough; it does not matter whether the blend is applied before or after differentiation.

5 Predictions and outcomes

6 What this means beyond the toy

- *The extraction works, and its accuracy is set by the puncture.* With an exact puncture the Taylor step is essentially free. With a truncated puncture it converges algebraically, at a rate

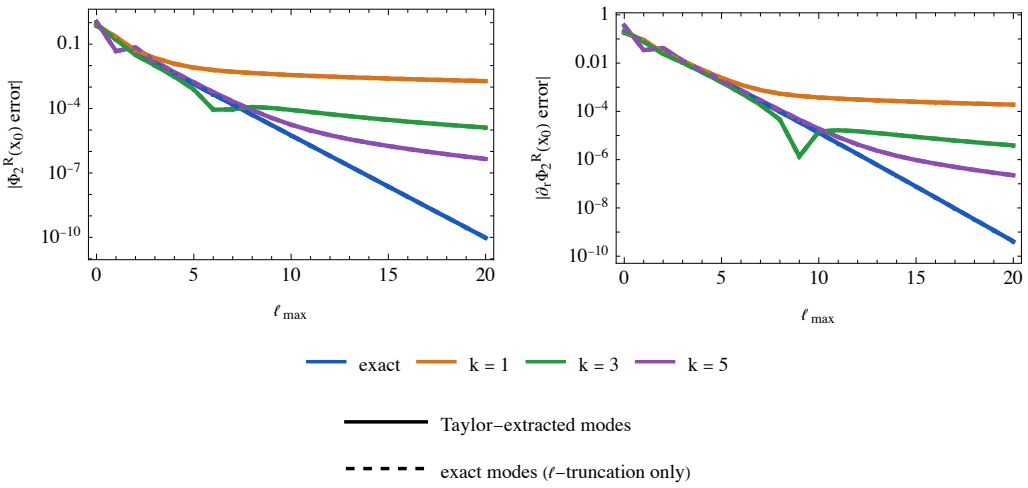


Figure 3: Error of the mode sums for $\Phi_2^{\mathrm{R}}(\mathbf{x}_0)$ (left) and $\partial_r\Phi_2^{\mathrm{R}}(\mathbf{x}_0)$ (right) against $\ell_{\max}$. Solid: Taylor-extracted modes; dashed: exact modes. The two coincide.

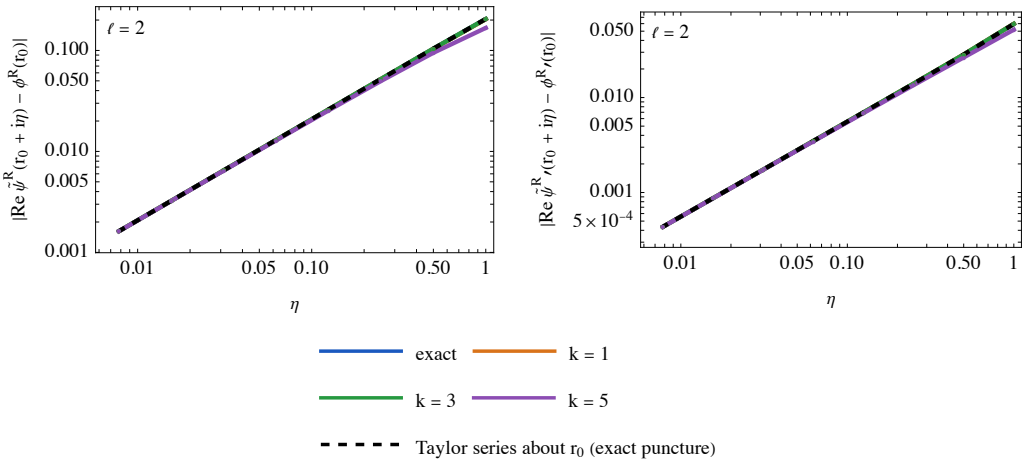


Figure 4: Baseline: bias of the residual read at $r_0+i\eta$ against η, $\ell = 2$, value (left) and derivative (right). Dashed: the Taylor series about r_0 for the exact puncture.

set by the smoothness of the residual *modes*, which is two orders better than that of the 3D residual. In either case the final mode sum is limited by ℓ-truncation, as in any puncture scheme.

- *What GR would need.* The Taylor route needs $\tilde{\psi}_\ell^{\mathrm{R}}$ and $\tilde{\psi}_\ell^{\mathrm{R}\prime}$ at one contour point, and the residual source $\tilde{s} - \mathcal{L}_\ell\tilde{\phi}_\ell^{\mathrm{P}}$ on one small circle about that point, which lies wholly in the upper half-plane. In GR the source there would come from the complex-r angular quadrature of Appendix C, which works at any point of the analytic region. The Stage 14 lesson applies: one set of source evaluations per point can be shared across all modes.

- *Two routes, similar reach.* The Taylor series and the η-extrapolation both reach 10^{-10} or better here. The Taylor route is cleaner and works from a single contour. The η route needs no source derivatives, but needs several contours and more nodes near the axis. For truncated punctures both are limited by the same branch point at r_0.

- *Precision and the shadow.* The shadow of the residual is of order one and grows with ℓ, to about 10 at $\ell = 20$. In double precision this sets the absolute accuracy of the extracted modes. That should be harmless for mode sums, whose terms are added absolutely, but it

prediction	outcome
Exact puncture: residual analytic at r_0; geometric convergence with ratio $\approx 1/5$ from the apex.	Held: ratio 0.199 against 0.196; monodromy $< 4 \times 10^{-20}$.
Truncated puncture: branch point at r_0; algebraic convergence $\sim N^{-k}$, worse for the derivative.	Algebraic held, but the exponent is $k+2$ for the value, because the mode functions are two orders smoother than the 3D field. The derivative is one order worse only off the vertical.
The puncture must be continued exactly as the source is, or the δ content is lost.	Did not hold: blending the finished, kinked puncture modes works. Dropping the kink fails.
Baseline loses about $2\log_{10}(1/\eta)$ digits to cancellation.	Did not hold: the field modes are only logarithmically singular. The limits are the $O(\eta)$ bias and the quadrature cost of the η^{-2} source.

limits the relative accuracy of high-ℓ modes.

- *Limits of the test.*
  - The toy is static, so it has no dissipative part. It tests the extraction of the local residual, which in GR carries the information that fluxes do not, such as the second-order redshift and binding energy.
  - The exact singular field is a finite closed form only because Φ_1^{R} is linear.
  - The residual was made nonzero for every ℓ by injecting a harmonic field.
  - GR adds m-modes, two-dimensional angular integrals, window functions and $\ln\rho$ terms in the puncture, none of which is tested here.

7 Reproducibility

The test is `stage15_residual_extraction.wls`, in the folder `Claude Implementation/Mathematica implementation/`. It runs in 22 minutes (3 minutes with the `smoke` flag) and passes all 363 of its checks. Results are in `results/stage15_*.csv`, with a detailed summary in `results/stage15_summary.md`. The figures, each with a CSV, are in the folder `Claude Implementation/figures/residual/`.